

Computer programing

Lab 11

Lab report

TITLE

Strings in C++ OBJECTIVE

The students will learn and apply:

- C-String
- Declaration and Initialization of C-String
- String Object

APPARATUS / SOFTWARE REQUIRED

- Computer System
- Dev-C++
- C++ Compiler

THEORY

- A string is a collection of characters. In C++, strings are used to store words and sentences.

There are two types of strings in C++:

1. C-Strings
2. String Objects **C-Strings:**

- C-Strings are character arrays terminated with null character \0.

Example:

```
char name[] = "Hello";
```

String Objects:

- String objects are provided by the C++ string class and can store long text dynamically.

LAB TASK 1 STRING LENGTH CALCULATION

PROGRAM

```
#include <iostream>
#include <cstring> using
namespace std;
    int
main()
{
    // Declaring string      char
    message[] = "Welcome to UET";

    // Finding length      int
    length = strlen(message);

    // Display output      cout << "Message:
" << message << endl;      cout << "Length =
" << length << endl;
    return
0; }
```

OUTPUT

Message: Welcome to UET Length
= 14

EXPLANATION

- strlen() function is used to find string length.
- Spaces are included in counting.
- Null character is not counted.

LAB TASK 2 STRING CONCATENATION

PROGRAM

```
#include <iostream>
#include <cstring> using
namespace std;
    int
main()
{
    // Declaring strings      char
    firstName[20] = "Aneesa ";      char
    lastName[] = "Ibrahim";

    // Concatenating strings
    strcat(firstName, lastName);
```

```
// Displaying result
cout << "Full Name: " << firstName << endl;

return 0; }
```

OUTPUT

Full Name: Aneesa Ibrahim

EXPLANATION

- strcat() joins two strings.
- Final result is stored in first string.

STRING INPUT USING cin.get()

PROGRAM

```
#include <iostream> using
namespace std;
int
main() {
    char str[100];
    cout << "Enter a string:
";    cin.get(str, 100);

    cout << "You entered: " << str << endl;

    return 0; }
```

OUTPUT

- Enter a string: Electrical
Engineering You entered:
Electrical Engineering

STRING OBJECT USING getline()

STRING OBJECT USING getline

PROGRAM

```
#include <iostream>
#include <string> using
namespace std;
int
main() {
    string str;

    cout << "Enter a string: ";
    getline(cin, str);

    cout << "You entered: " << str << endl;

    return 0;
}
```

OUTPUT

- Enter a string: University of Engineering and Technology You entered: University of Engineering and Technology

RESULT

- The programs were executed successfully and the concepts of C-Strings and String Objects were understood.

CONCLUSION

In this lab, we learned:

- C-Strings
- String Objects
- String length calculation
- String concatenation
- Using cin.get() and getline() for input